

Samridh Prabhakar

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EDUCATION

Purdue University

B.S. Computer Engineering, Certificate in Semiconductors and Microelectronics

Graduating 2028

GPA: 3.36

EXPERIENCE

Undergraduate Researcher

Purdue Vertically Integrated Projects, Semiconductors @ Birck

Jan 2026 – Present

West Lafayette, IN

- Fabricated 6 wafers of Ti/Al-HfO₂-Si **MIM capacitors** through a full photolithography process flow, targeting 1800 nm AZ1518 resist thickness and 0.28 mm² electrode geometry.
- Operated a **Filmetrics F40-UV** thin-film analyzer to measure photoresist thickness via spectral reflectometry across wafers, verifying profiles remained within the defined process window.
- Made **SPC control charts** for AZ1518 film thickness and capacitor top electrode area across a 6-wafer run.

Student Researcher

Purdue Vertically Integrated Projects, Virtual Twins @ Birck

Aug 2025 – Dec 2025

West Lafayette, IN

- Modeled semiconductor fabrication equipment as virtual twins for visualization on the **Apple Vision Pro**, supporting cleanroom operator training in a VR environment.
- Performed motion analysis and kinematic validation of PVD equipment assemblies in **Dassault Systèmes CATIA**, producing mechanical drawings used for design review.

Research Intern

University of Delhi

Apr 2023 – Jan 2024

Delhi, India

- Ran specification based electrical tests of **silicon microstrip sensors** for the CMS Phase-II Upgrade at CERN, characterizing leakage current, capacitance, and resistance to validate sensor reliability.
- Flagged defective sensors against performance thresholds, directly supporting quality acceptance criteria for detector components destined for LHC installation.

PROJECTS

TidalGate | C++, Python, Arduino Uno Q | DataHacks 2026 Winner

- Integrated a Modulino IMU, Thermo module, and analog depth sensor onto an **Arduino Uno Q**, fusing 3D motion, temperature, and wave height data into a real-time tidal energy estimation pipeline.
- Wrote the data bridge between the embedded sensor layer and an on-device **Qualcomm-hosted Python ML model**, handling serialization and synchronization across heterogeneous platforms.

Embedded Tetris System | C++, ESP32, SPI, ESP-NOW

- Designed and developed a handheld **ESP32-based embedded gaming device** to run a real-time Tetris game.
- Built a graphical interface on an SPI-driven display, rendering game state, score, and next-piece preview
- Integrated a heuristic piece ranking engine to dynamically select the worst possible next piece for the player, adapting the difficulty in real time

Location Based Digital Compass | Python, Raspberry Pi Zero W, Google Maps API

- Built a handheld navigation device that resolves the nearest point of interest using real-time GPS, compass heading, and the **Google Places API**, displaying destination name, bearing, and distance on an OLED screen.
- Implemented Wi-Fi geolocation fallback and API response caching to maintain reliable operation in GPS-degraded environments.

SKILLS

Programming: C, C++, Python, MATLAB, Verilog

Cleanroom Skills: Photolithography, Thin Film Measurement, ALD, Mask Alignment, SPC

Hardware & Systems: Embedded Systems, PCB Design (KiCad), CATIA, Oscilloscopes

Development Tools: Git, VS Code, PlatformIO, Google Cloud Platform

Certifications: Linux and Bash (Purdue IEEE), Semiconductor Manufacturing 101 (Purdue, UTA & Intel), MATLAB Onramp (MathWorks)